

THE EMERGENT BELT · Vision 2035



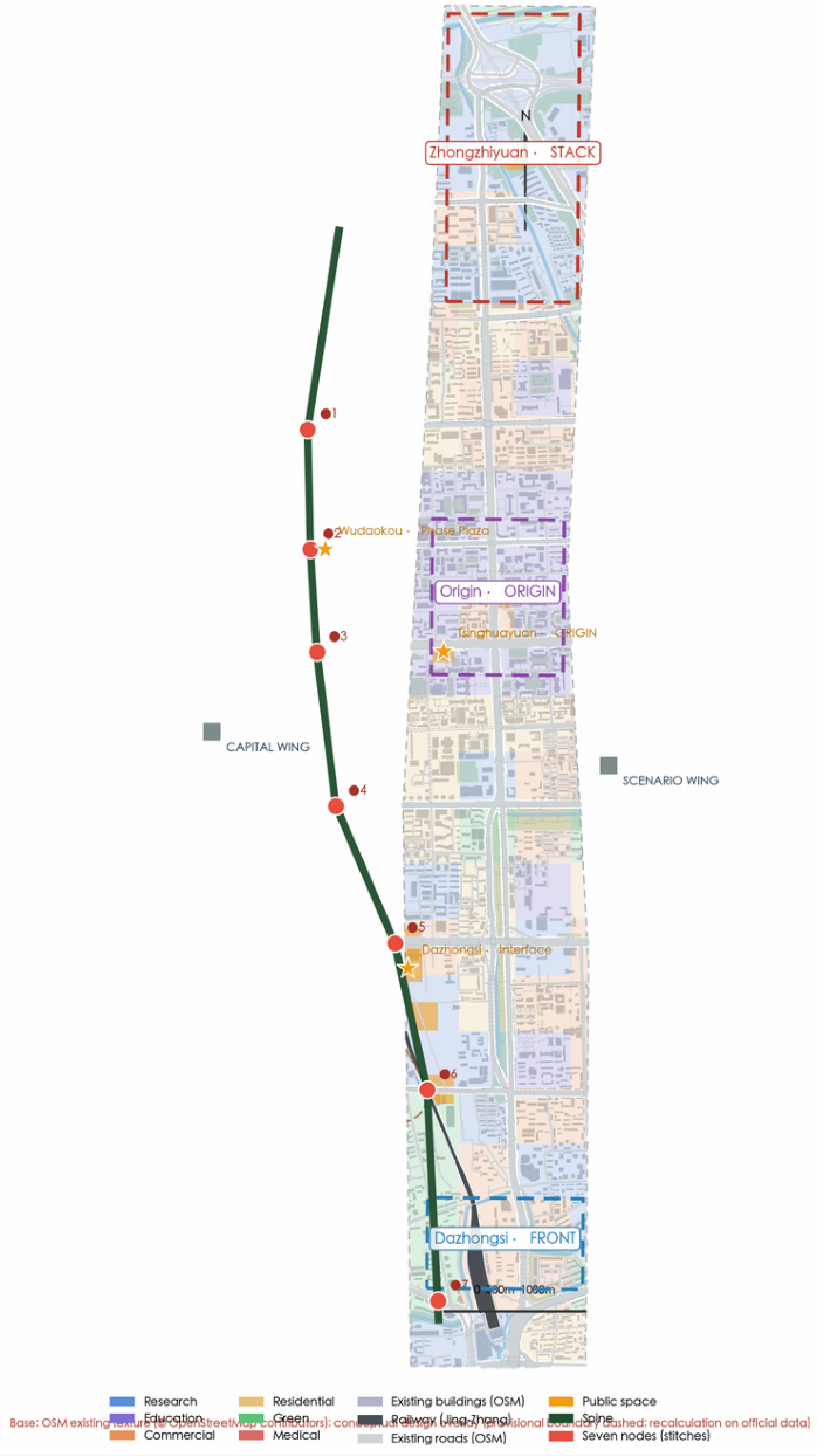
Space is not drawn — it is computed

The century-old corridor emerges into an AI belt. Method: CAS+percolation+syntax+LUTI, all computed.

Philosophy: mosaic retrofit + toolbox planning — 85/15 reserve. Spine as symphony: 3 identities.

Master plan (7.9km spine · 3 areas 2 wings 7 nodes)

Fig.1 · Master Plan: one line, three folds, two wings, seven nodes (concept)



Spine night



Block dusk



Stitch bridge



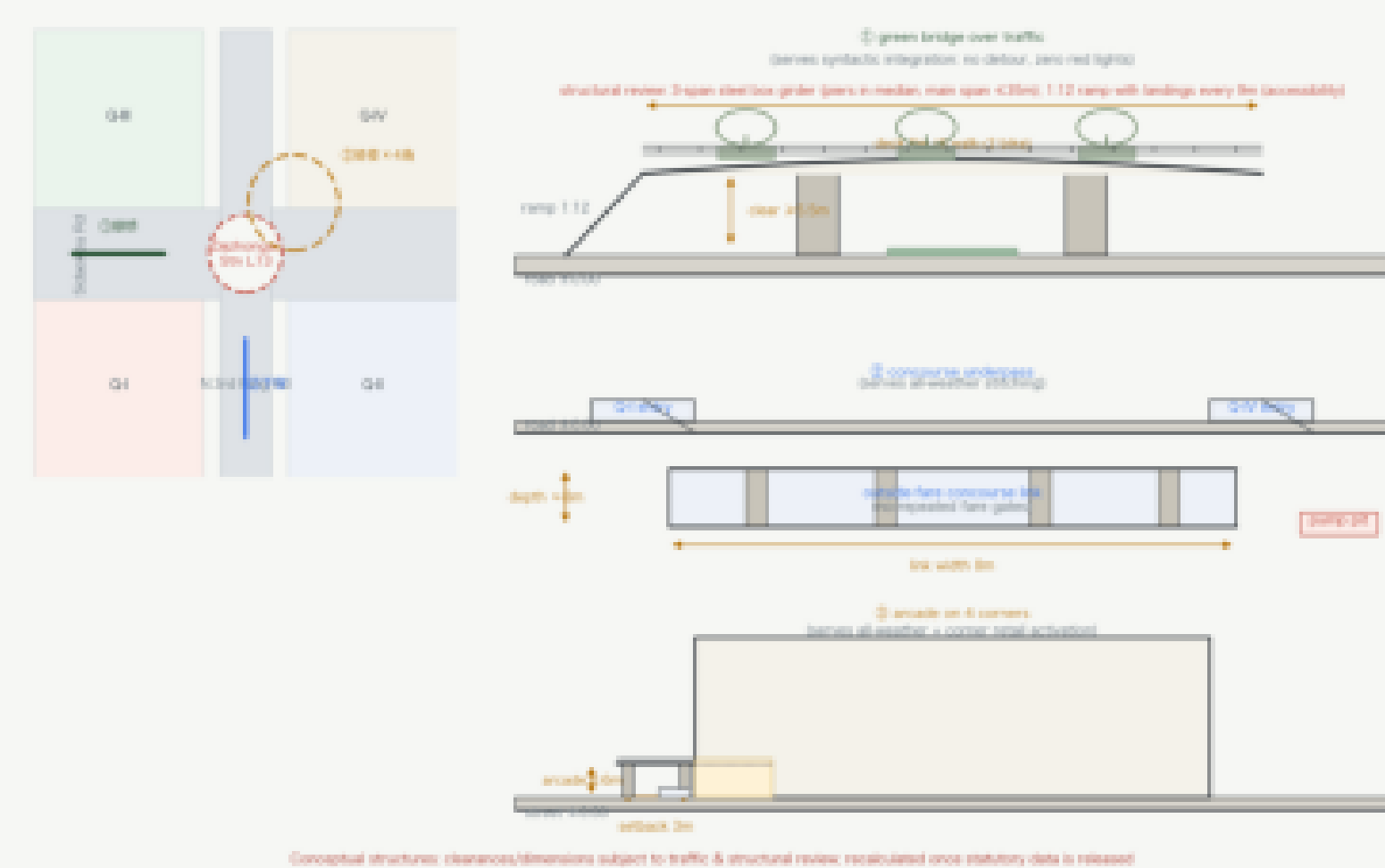
Dazhongsi front



7.9km	3	7	0.904	+36%	+5.3%	28
spine	world-first	stitches	percolation	syntax	entropy	local metrics

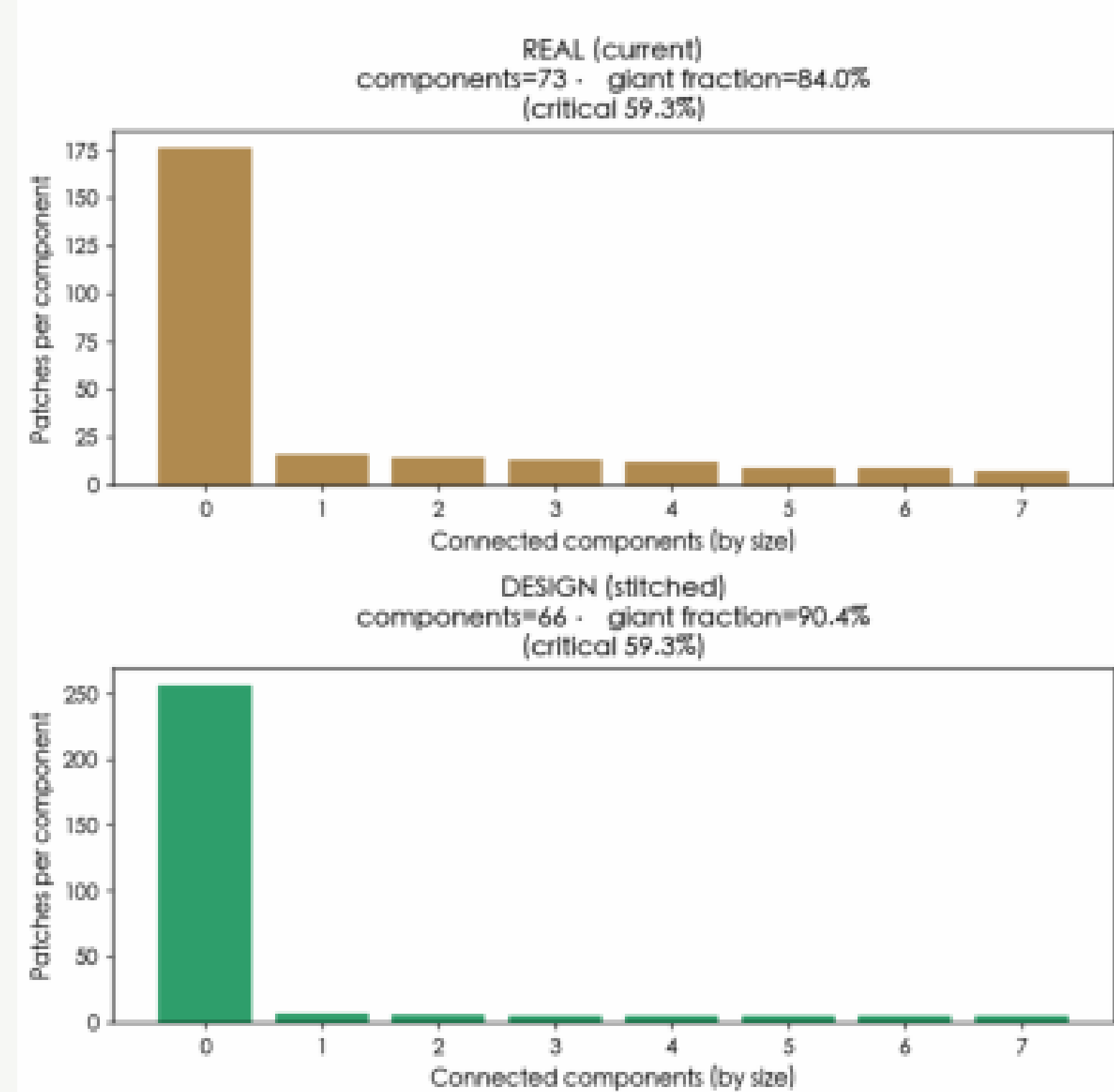
Dazhongsi four-quadrant · stitching structure design (concept, not construction drawings)

Anchor: W 3rd Flg. K. Solitaires crossing. Tree stitches + green bridge + concourse underpass + arcade selflink (all serve syntactic integration & all-weather passage).



Percolation (giant 0.904)

Fig C2 • Blue-green percolation: current vs design stitching

[illegible]

Syntax gain +36%

Fig C4b - Toledo-style space syntax:
configuration shapes movement (Jing-Zhang)

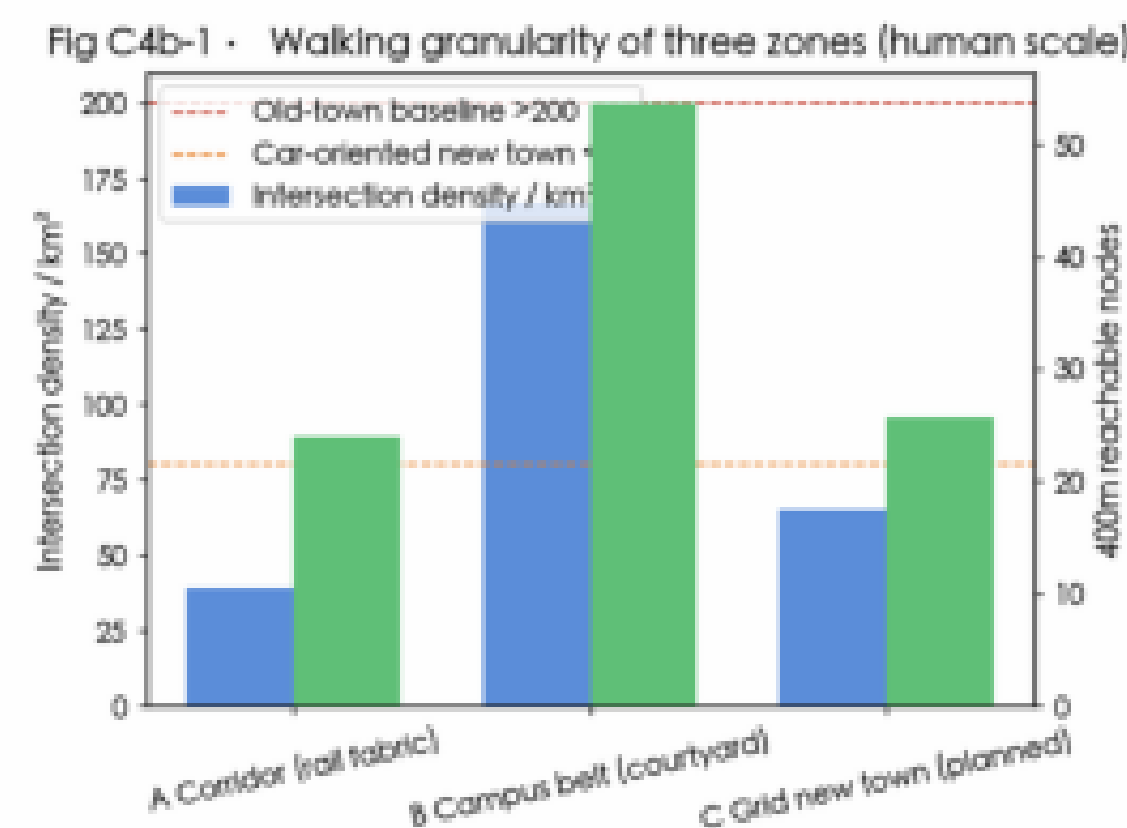
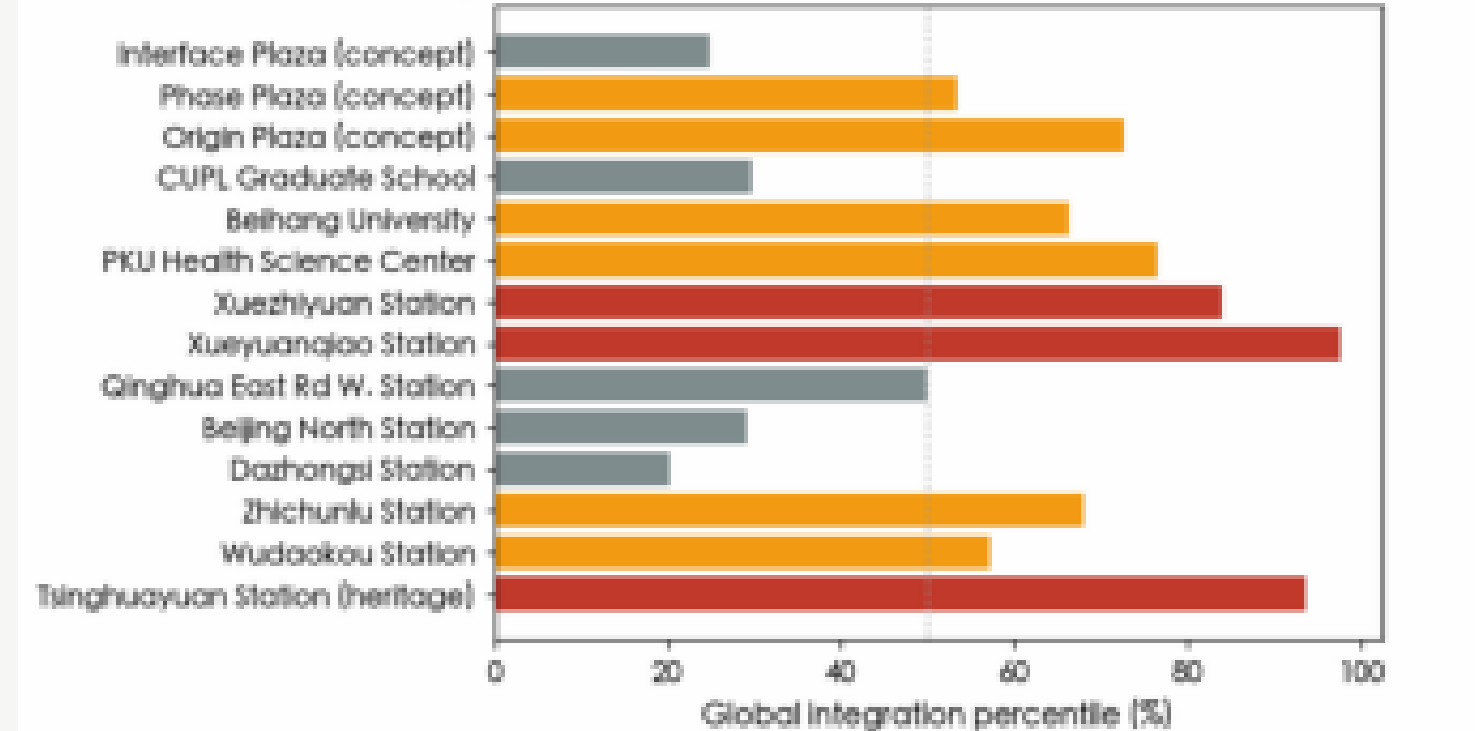


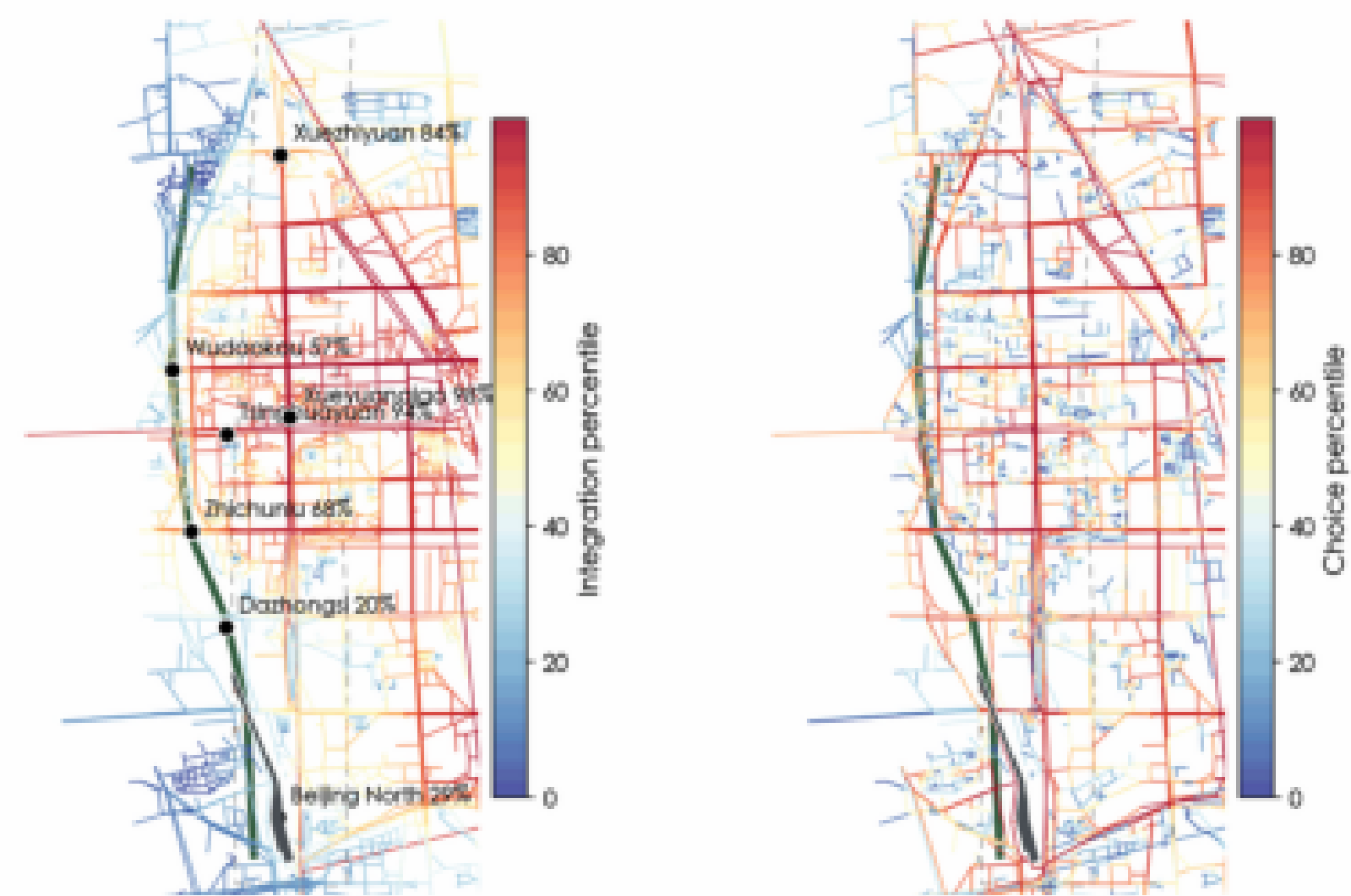
Fig C4b-2 • POI syntax: who is Jing-Zhang's Bisagra Gate?



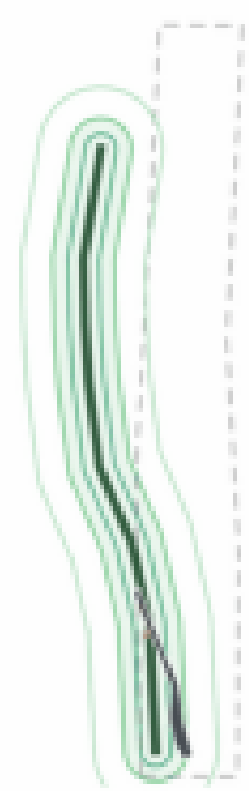
Syntax integration

Space syntax plan analysis: configuration shapes movement (5.720 nodes, computed)

Map A · Integration: red = structural core Map B · Choice: red = through-movement spine



Map C · 200/400/800m walking rings: the hardest 400m on the belt



Stitch bridge



Spine night



Three gates (all met)

1 Giant ≥ 0.90 : 0.9044

2 Syntax $\geq 30\%$. +36%

3 Entropy $\geq 4\%$: +5.3%

7

stitch nodes

2

bridges

 $\lambda=0.42$ critical λ

0.904

glant

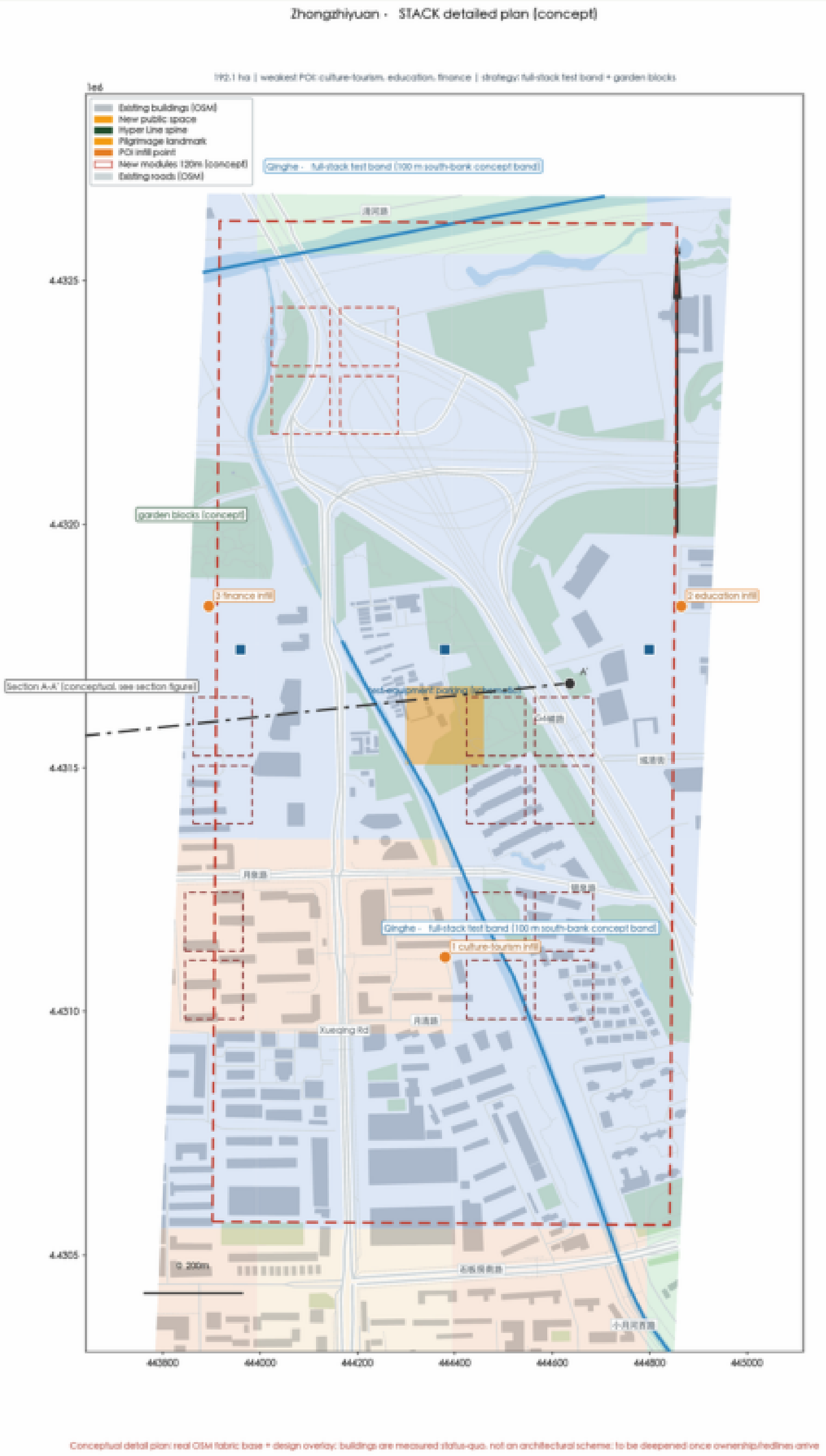
+36%

syntax

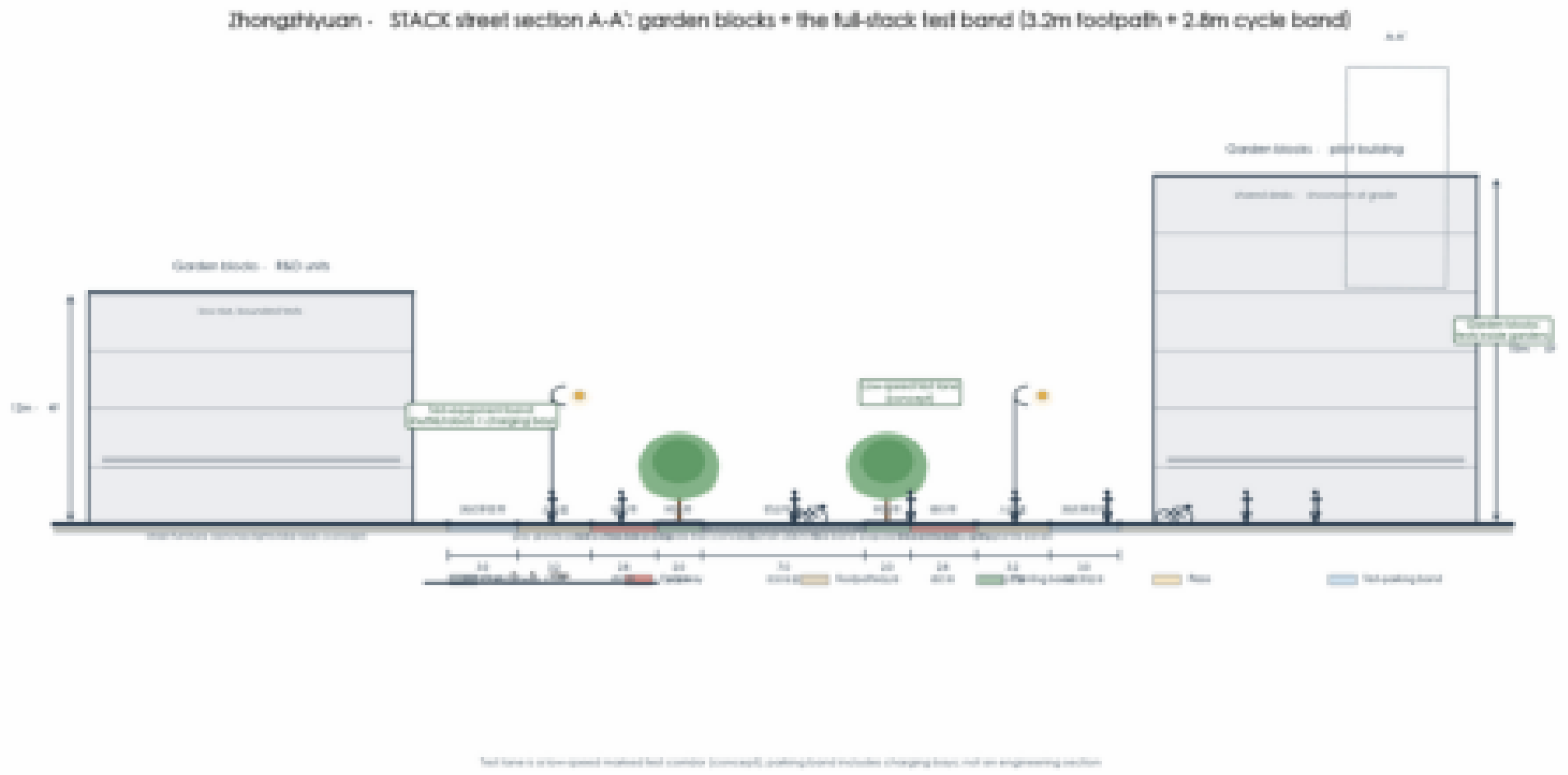
15/16

same-comp

I Zhongzhiyuan



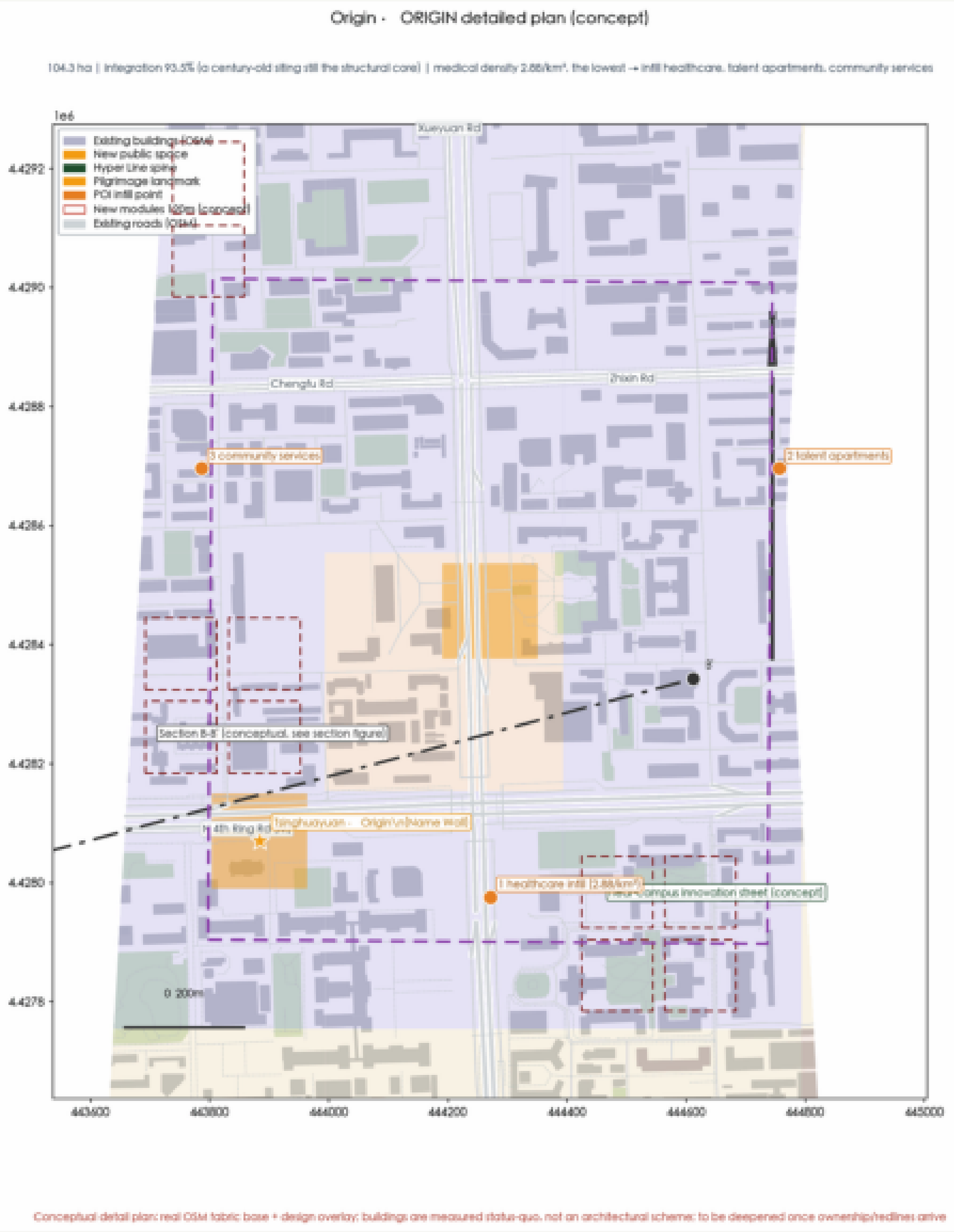
Section · mix



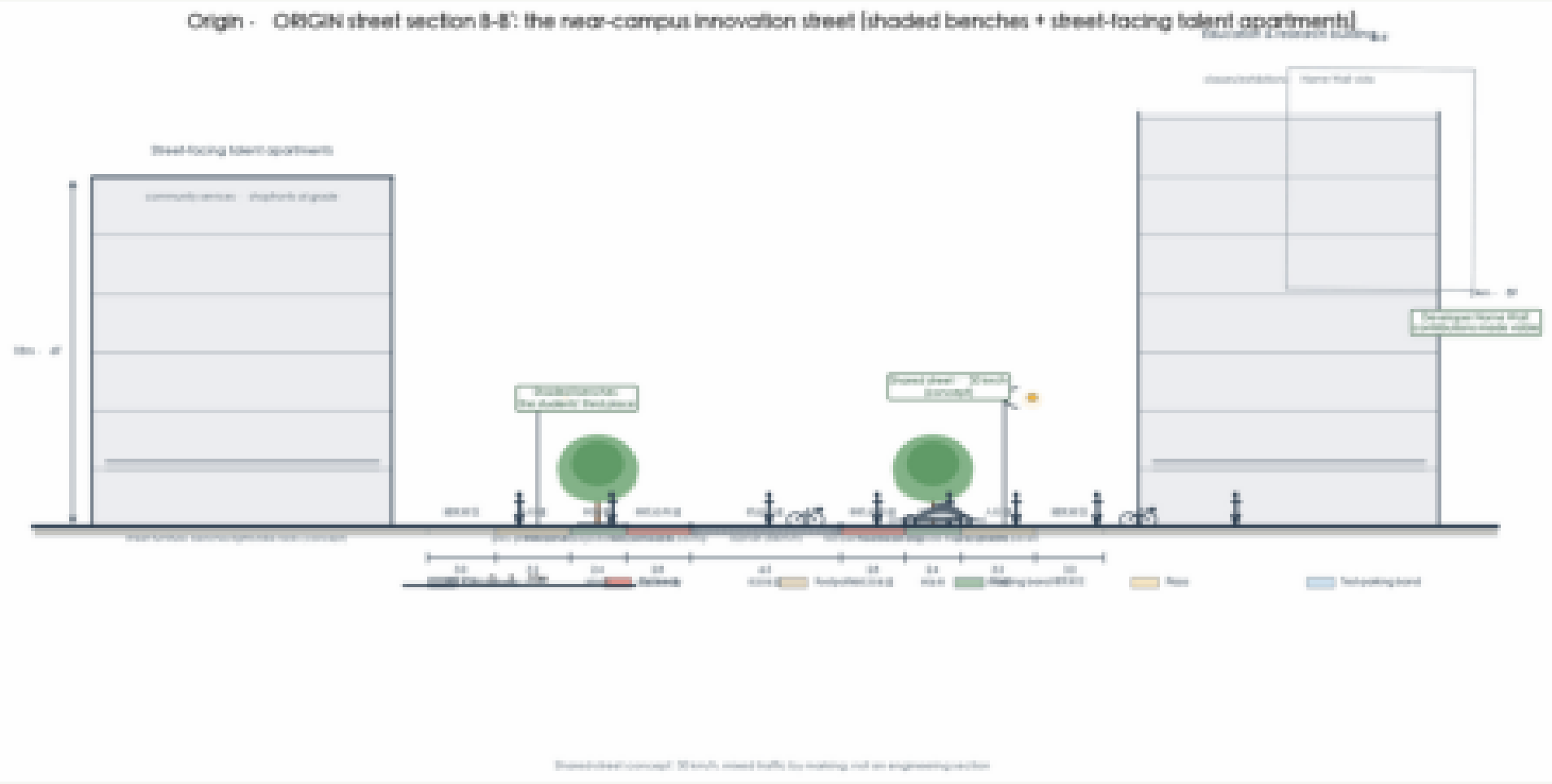
Shared workspace



II Origin Community



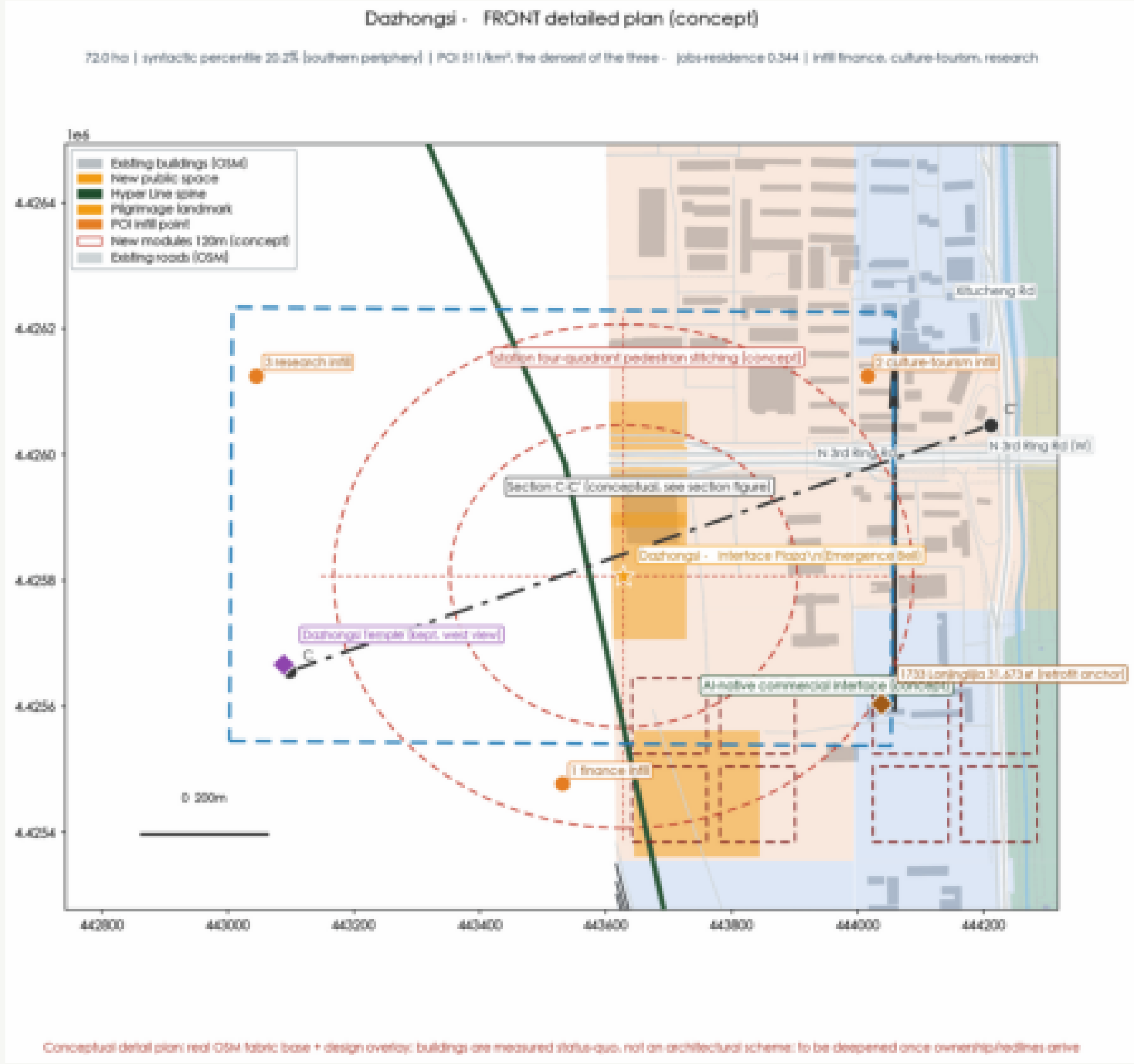
Section · mix



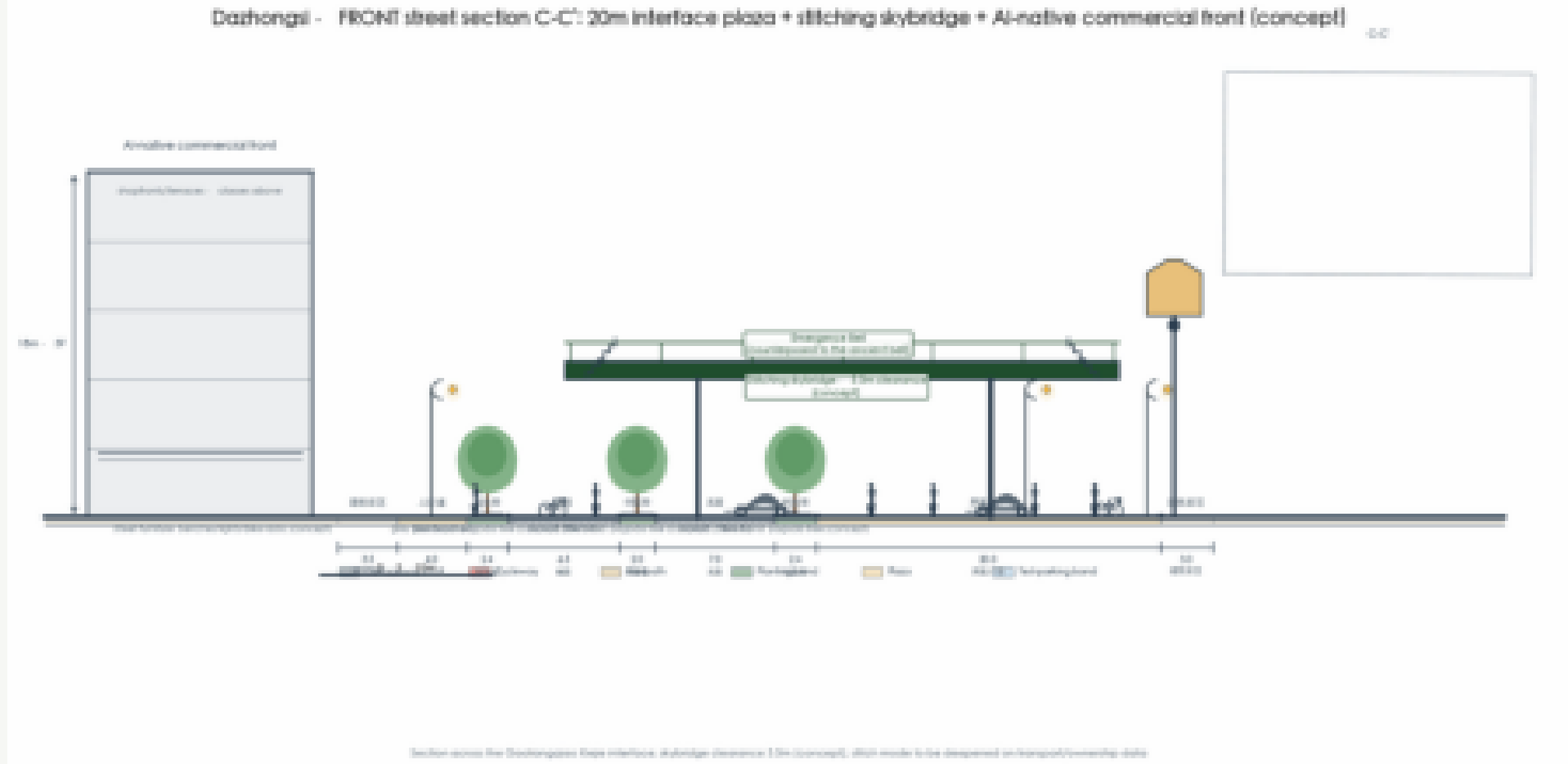
Origin community



III Dazhongsi



Section · mix



Phase plaza



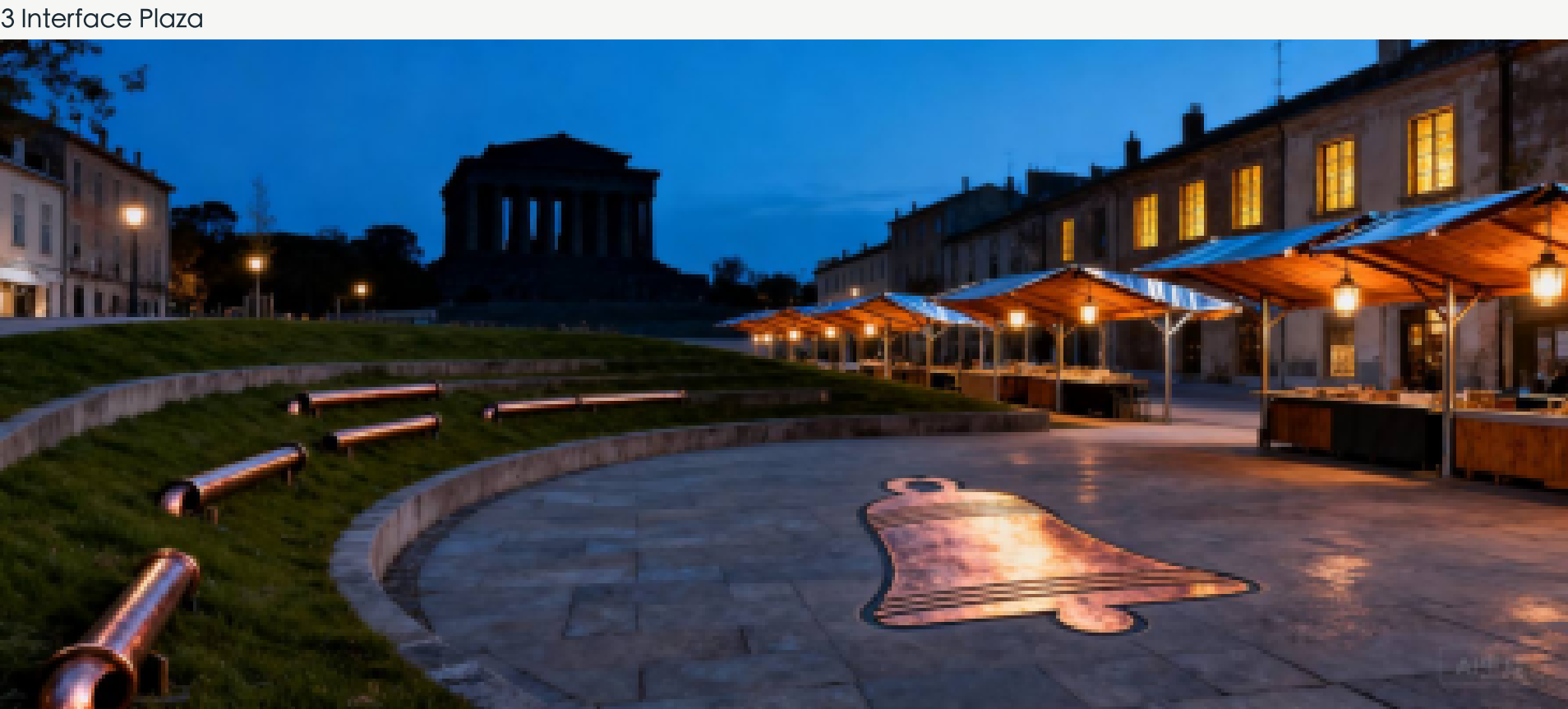
3	85/15	44.6万㎡	FAR0.91	141/km²	3
key areas	reserve	new build	FAR	job density*	identities



85/15: design 85%, leave 15%

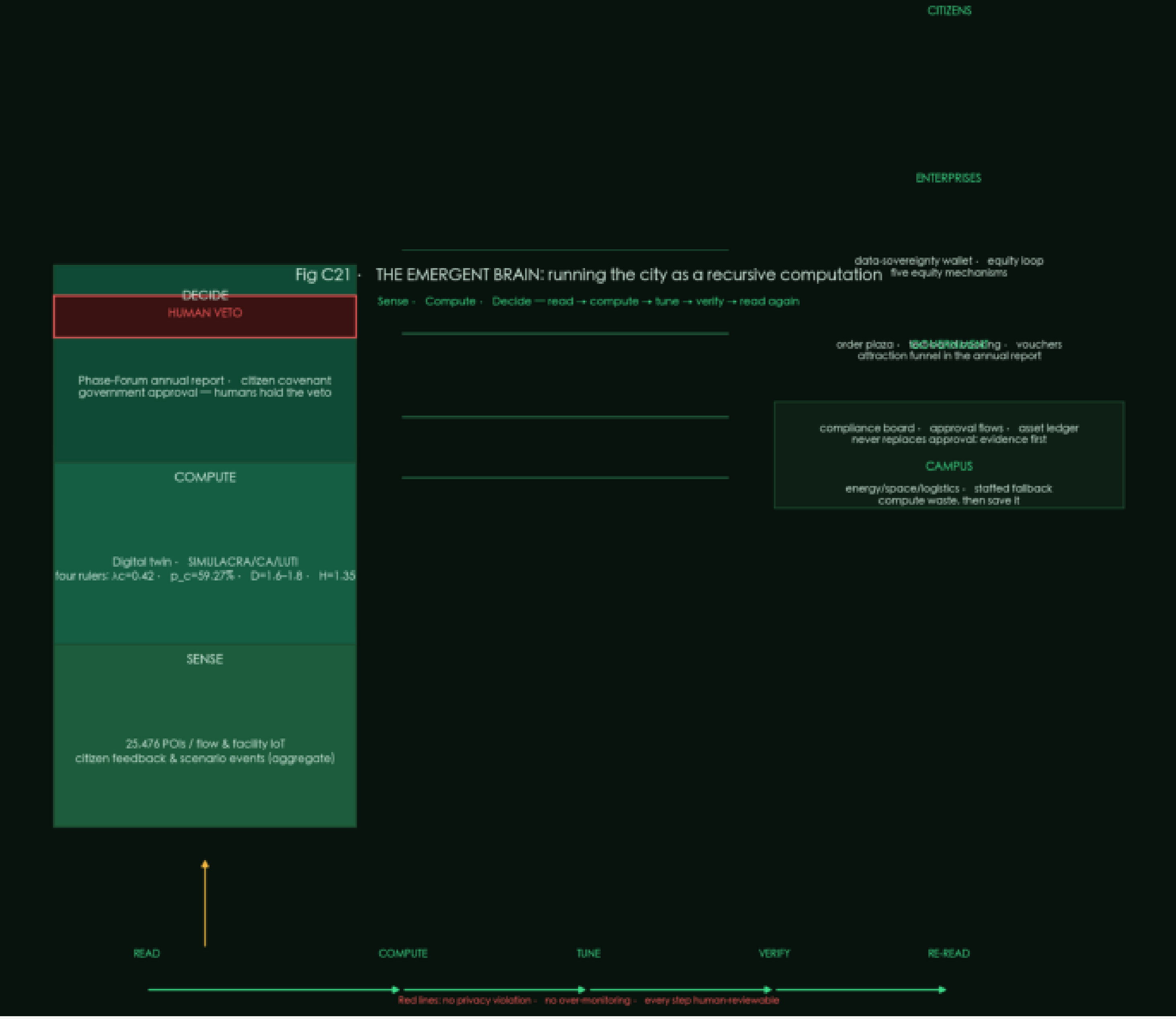


1909 origin · heritage as living room

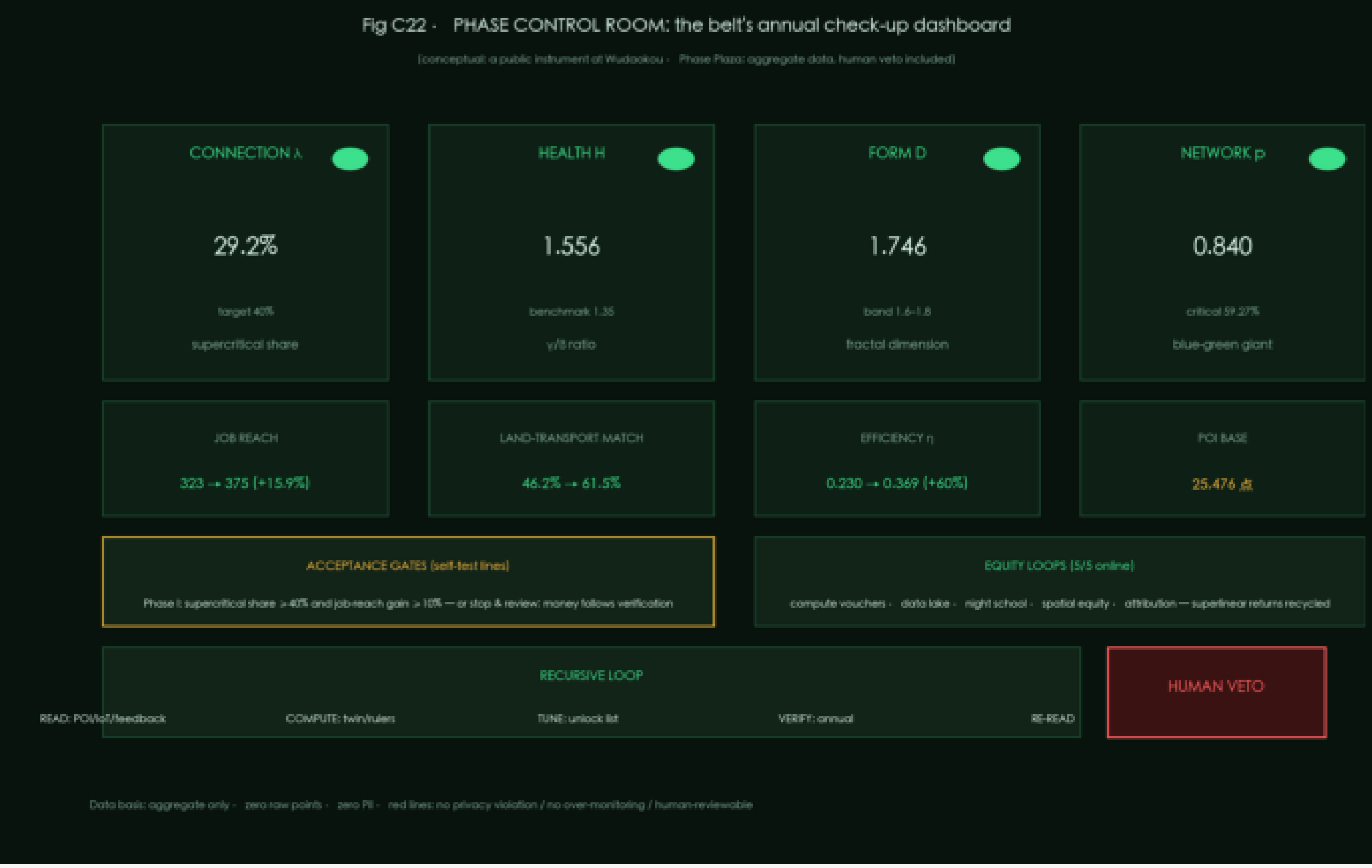


Emergence Bell+1733 Interface

Emergent Brain



Phase control room



3	4	5/5	1733	85/15	1909
layers	terminals	equity	interface	reserve	heritage

CA generation

Fig C12 - CA urban-renewal emergence simulation: no intervention = lock-in, intervention = unlock (computed)

Initial (POI status quo, bottom-up)
693 cells

No intervention, 60 steps (path lock-in)
0 cells changed

Intervention, 60 steps (stitching + renewal unlock)
42 cells changed

Overlap with the manual 400m zoning 38% (emergence under constraints vs imposed order)

Figure C1 · Wilson spatial-interaction λ calibration (status quo)

$\lambda_{ij} = \exp(-d_{ij}/2500m)$; $\lambda > \lambda_c = 0.42 = \text{reachable within the 10-min innovation circle (DSM network)}$

Ten core pairs: only one crosses the line (Origin x Xiaoyuehe)

Core Pairs	Coupling λ_{ij} (network basis)
北京AI原点社区 x 小月河场景赋能	0.42
北京AI原点社区 x 中关村科技服务翼	0.374
中关村科技服务翼 x 小月河场景赋能	0.296
大钟寺AI产业集聚区 x 小月河场景赋能	0.223
大钟寺AI产业集聚区 x 中关村科技服务翼	0.222
众智园AI自主创新加速区 x 北京AI原点社区	0.205
北京AI原点社区 x 大钟寺AI产业集聚区	0.162
众智园AI自主创新加速区 x 小月河场景赋能	0.093
众智园AI自主创新加速区 x 中关村科技服务翼	0.091
众智园AI自主创新加速区 x 大钟寺AI产业集聚区	0.033

Coupling λ_{ij} (network basis)

Supercritical share · euclidean: 50.00%

Supercritical share · network: 29.17%

5-core global connectivity: No (severed)

Diagnosis: the five cores are not connected under the 10-min coupling — a healthy skeleton severed by seven roads (λ is cut one of the diagnosis ring).

Fig C15 • LUTI evaluation: jobs-housing balance $\times$ land-transport match $\times$ commute flows (Jing-Zhang)

LUTI mismatch map + commute flows
green=underdev · red=overdense · gray=matched
(current matched 46%)

Land-transport match (LUTI loop closure)

Scenario	Matched cells (%)
Now	~46
After stitch	~61

Jobs-housing balance (mean 0.39)

Scenario	Cells with JHR in [0.4, 0.6] (%)
Now	~38

Healthy band target 60%

Fractal dimension

Fig. C13 · Fractal spectrum (area-perimeter / density decay / lacunarity)

The figure consists of three bar charts. The first chart, titled 'Area-perimeter D', compares the fractal dimension of buildings (blue bar, value 1.04) and a manual 400m grid (green bar, value 0.58). The second chart, titled 'Employment density decay γ ', compares the fractal dimension from stations (blue bar, value 1.8544565685184056) and from the spine (green bar, value 1.8544565685184056). The third chart, titled 'Green texture $A=5.072832204534368$ ', shows the fractal dimension of green texture (orange bar, value 5.072832204534368).

Category	Fractal Dimension
Buildings	1.04
Manual 400m grid	0.58
from stations	1.8544565685184056
from spine	1.8544565685184056
Green texture	5.072832204534368

Fig C20 - Investment budget and return estimation (conceptual magnitude, not an investment commitment)

A - Investment breakdown: 16.98-30.89 in total (mid 23.9)

Category	Value Range (CNY 100M)	Mid-Point Estimate (CNY 100M)
Seven-node stitching	1.21-1.28	1.24
Public space	1.20-2.40	1.80
Interface renewal	0.50-1.00	0.75
Talent apartments	4.50-7.40	6.00
Test band & gardens	1.47-2.33	1.90
Emergent brain platform	3.30-4.00	3.65
Operations fund	0.00-0.00	0.00

B - Market supply-demand ledger (conceptual)

Demand side (computed in this proposal)

- Work-type PCs: 4919 (companies+research+finance)
- 13,000 developers + 180 PCs + 1,000+ scientists
- Jobs+housing balanced cells: only 38.4%
- Office increment = 100k m² (research gap -45.9%)
- 2,000 talent apartments (conceptual infl)

Supply side (conceptual rent references)

- Office/hotel: 3.5 CNY/m²/day × 100k m² × 75%
- Apartments: 2,000 CNY/month × 2,000 × 85%
- Steady gross = 1.67 → net cash flow = 1.17 /year

Return paths (three ledgers)

- Asset value: y10 net recovery 1.40n (RETI/diaposol, conceptual)
- Social ledger: reach = 15.9% / jobs+housing / H1.56 (not discounted)

C - 20-year cash flow and payback (concept)

D - Sensitivity: rent/occupancy/capex ±20%

The dashboard is titled "Design conclusion dashboard: four sentences, each with numbers" and is organized into four main sections, each with a numbered title and three data points.

- Section 1: Not rebuilding, but re-growing**
 - D = 1.746 (fractal dim - band 1.6-1.8)
 - H = 1.556 (health factor - benchmark 1.35)
 - 0.840 (blue-green giant - critical 59.27%)
- Section 2: Move aside what blocks emergence**
 - 7 → 7 (gaps → stitches)
 - 42 (cells unlocked [CA])
 - 3 (middle-layer indicators annualized)
- Section 3: Falsifiable, cross-verified**
 - +15.9% (25-min job reach)
 - +15.3pct (match rate 46.2% → 61.5%)
 - +60% (efficiency η - Nash D.D. → C.C.)
- Section 4: An operating system, not a blueprint**
 - 0.42/59.27% (A/C / p,c critical)
 - 1.6-1.8/1.35 (D band / H benchmark)
 - 1台 (Emergent brain - check-up - cell)

A footer section titled "Ten-year timeline - two acceptance gates" includes a timeline from 1-3y to 5-10y, with specific goals for each period: stitching + Daohongji (1-3y), Origin renewal (3-5y), and gates: 1.0-40% / H = 1.35 (5-10y), with a note "Zhongjiyuan gardens".

The dashboard concludes with the statement: "The future cannot be predicted, but it can be invented. — Michael Batty".